*Exploring the World of Science*

University of Michigan Science Olympiad
2022 Invitational Tournament

# Dynamic Planet C

**Test length:** 50 Minutes

**Team name: Key**

**Student names: Key**

Notes: There are a total of 127 points spread out over 12 Stations, all questions have their point value listed by the answer. Answers are underlined or written in for multiple choice or fill in answers respectively. Bolded answers on multiple choice questions are worth fewer points.

**Tie Breaker Questions (All Labeled on Key)**

- 10, 12, part of 32, and 39

1. What is the proper cross sectional shape of a river valley? (1)
    a. U-shaped
    b. V-shaped
    c. Rectangular
    d. Flat
    e. L-shaped
2. What causes the formation of lakes? (1)
    a. The rifting of continental crust leading to deeply slipped grabens
    b. The damming of rivers by glaciation or sedimentation
    c. The collapse of subsurface magma chambers after erupting
    d. All of the above
    e. A and B
3. A braided stream environment can be characterized by (steeper/shallower) gradient and (coarser/finer) sediment than meandering streams. (2)
4. What is a more characteristic feature of meandering streams? (1)
    a. Oxbow lakes
    b. Intermedial lobes
    c. Alluvial fans
    d. V-shaped beds
    e. Carbonate deposits
5. Old streams tend to have _____ beds with _____ competence. (2) (Bold = 1 pt)
    a. Narrow, v-shaped; low
    **b. Wide, flat; high**
    c. Wide, flat; low
    d. Narrow, flat; high
    **e. Wide, v-shaped; low**
6. A knickpoint in a stream could be characterized by: (1)
    a. Rapids
    b. Waterfalls
    c. Eddies
    d. Deltas

e. Islands

7. Discharge in a stream tends to increase after precipitation, though some experience greater time lag than others. An increase in peak discharge time lag can be caused by: (1)
    a. Low vegetation
    b. Frequent rainfall
    c. Low permeability soil
    d. Channelization
    e. Groundwater depletion
8. True or False: Radial and annular drainage patterns are more common on volcanic islands. (1)
9. Alluvial fans form when there is a sudden change in _gradient_ and tend to have _intermittent/seasonal_ deposition and flow. (2)
10. What dissolved species might not be prominent in a stream that runs over igneous rock? (1)
    a. $HCO_3^-$
    b. $K^+$
    c. $H_4SiO_4$
    d. $Na^+$
    e. $Ca^{2+}$
11. A 100-year flood: (1)
    a. Has a 1% chance to occur each year
    b. Occurs on repeating 100 year cycles
    c. Often destroys or alters the course of a stream
    d. All of the above
    e. A and C
12. Conglomerate is a rock made of large sized clasts with fine grained matrix or mineral cement. If you found a narrow, thin bed of conglomerate with fine grained cement, you could assume that: (1)
    a. The capacity of the depositing stream was high
    b. The sediments are mature
    c. The depositing stream likely had a high velocity

d. The stream had a low gradient
e. B and C

13. A clay particle is anything that has a diameter of less than: (1)
    a. .125 mm
    b. .0625 mm
    c. .0156 mm
    d. .0039 mm
    e. .00006 mm
14. A graph that shows the discharge of a river over a given time is called a: (1)
    a. Hydrocline
    b. Hypsometric curve
    c. Halocline
    d. Hydrotherm
    e. Hydrograph
15. _Clay_ minerals tend form as the weathering products of silicate bedrock and produce very _fine_ clasts. (2)
16. You want to track the flow of water in a river. This can often be accomplished by adding _____ to the water and measuring when it reaches a downstream point. (1)
    a. Salt
    b. Dye
    c. Alkaline solution
    d. Acidic solution
    e. Charcoal particles
17. The _center surface/thalweg_ is the fastest moving part of the stream. (1)
18. Which of the following would not be an effective aquitard? (1)
    a. Shale
    b. Granite
    c. Rhyolite
    d. Limestone
    e. Basalt

19. True or False: A pH change from 10 to 6 in a body of groundwater is anomalously high. (1)
20. True or False: Groundwater always flows downhill. (1)
21. Saltwater intrusion can be caused by _____ of aquifers near _____. (2)
    a. Pollution; cities
    **b. Overpumping; evaporites**
    c. Discharge; marine coasts
    d. Overpumping; marine coasts
    e. Discharge; evaporites
22. Artesian wells are the term given for: (1)
    a. Naturally occurring wells of any water at the surface
    b. Wells pumped from confined aquifers that reach the water table
    c. Wells pumped from confined aquifers that reach the piezometric surface
    d. Wells pumped from unconfined aquifers that reach the piezometric surface
    e. Wells that flow from joints present in an aquitard
23. Which of the following can cause groundwater depletion? (1)
    a. Overdevelopment of watershed surface
    b. Change in water composition
    c. Channelization of waterways
    d. Increased sediment supply in waterways
    e. A and C
24. Deltas tend to be fertile agricultural zones due to their extensive deposits of _silt_, rich in nutrients. (1)
25. Which of the following is NOT true about deltas? (1)
    a. They are almost universally a fan-like structure of sediments and distributaries
    b. They provide important wetland habitat for various species
    c. They are at severe risk of inundation by climate change
    d. They more often than not are at the end of meandering, low gradient streams
    e. Their ancient sedimentary sequences often include coal
26. Karst topography would be least likely to occur in a location with which bedrock? (1)
    a. Limestone

b. Gypsum
c. Dolostone
d. Shale
e. Rock Salt

27. A stream that terminates before reaching a surface body of water is a: (1)
    a. Misfit stream
    b. Karst window
    c. Gully
    d. Disappearing stream
    e. Dissolution valley
28. Soils in karst regions tend to be (acidic/neutral/basic). (1)
29. Where does Karst topography predominantly occur in the US? (1)
    a. New England
    b. Midwest
    c. South
    d. Southwest
    e. Pacific Northwest
30. Tower karsts tend to be indicative of (older/younger) stages of weathering.
31. See Image Set I; Buffalo City, AR Quadrangle. (Ask the supervisor if you have trouble reading the names!)

What is the name of the main channel in this quadrangle? What direction is it flowing? What is the drainage pattern of the system? What is the Strahler stream order of streams A, B, and C? Would you expect this river to be younger or older? Why? (8)

White River; West to East; Dendritic; 1, 2, 2; Older due to wider bed (1), low gradient (1).

32. See Image Set II; Twin Falls, ID Quadrangle

Circle all of the knick points visible on this map. What is the type of cross section of this river? How might this have formed? **TB:** What do the blue dots with squiggly tails on either side of the river indicate? (7)

(4, 1 each for any 4); Rectangular (1); Glacial dam bursting (1) carved out the valley in a highly pressurized and destructive flood (1); Springs/Seeps (1 TB point).

33. See Image Set III; Lake Mary and Brandreth Lake Quadrangles

One of these quadrangles is from a region of karst topography while the other is glacialized terrain. Which is which? What are at least two trends or details that support your choices? (6)

Lake Mary is karst, Brandreth is glacialized (2); (1 each for any point made) Brandreth has connected waterways, Mary does not; Mary has small rounded ponds, Brandreth has more elongate lakes; Brandreth has more dramatic topography with few depressions, Mary has many sinkholes and is flatter. Brandreth topography has a direction to it, Mary does not.

34. Match each present or past lake to its corresponding type. More than one lake can correspond to each type (8)

Lake Baikal, Russia (Rift)

Crater Lake, Oregon (Volcano)

Lake Michigan, Midwest (Glacier)

Lake Superior, Midwest (Rift)

Lake Bonneville, Utah (Dam)

Lake Missoula, Idaho/Montana (Dam)

Lake Malawi, East Africa (Rift)

Lake Geneva, Switzerland/France (Glacier)

Volcanic Depression

Glaciation

Rift

Glacial/Sediment Dam

35. What is the difference between capacity and competence? (2)

Capacity is the amount of sediment that can be held by a fluid (1); competence is the largest size of sediment that a fluid can carry (1)

36. What are the two main factors of a fluid that control sediment transport and deposition? (2)

Flow velocity (1) and flow depth (1)

37. What does Darcy's Law describe? What would happen to the described property if the dynamic viscosity of water went up? What would most likely cause the viscosity of water to increase? (4)

The instantaneous flow rate of a fluid (1) through a porous medium (1); the flow rate would decrease (1); a drop in temperature (1)

38. What forms the boundary between the zone of aeration and the zone of saturation? What is an aquifer called when it is confined but under this boundary? How might this boundary be "perched" in some places? (5)

Water table (2); Artesian (1); a piece of impermeable bedrock sits above more contiguous or large aquitards and the water accumulates on top of it, forming a localized water table. (2)

39. **TB:** Many mining regulations are determined by the watershed that an operation is located in. Why would this be the case? What are some environmental hazards posed by mining? (5)

Watersheds are the area in which water drains into streams, so pollutants in runoff or groundwater will eventually flow into the streams (2); acid rock drainage, sedimentation, heavy metals, stream diversion, processing contamination (e.g. cyanide) (3, 1 each for any 3)

40. Why are streams channelized and what are some of the consequences of this? (4)

Preventing floods in major areas by increasing flow velocity (2); can increase erosion and sedimentation downstream, easier pollutant runoff (2)

41. What main factors control the density of water? How does each affect it? (4)

Temperature (1); decreases with density (1). Salinity/dissolved ions (1); increase with density (1).

42. What element is notably absent in swamp and marsh soils? Why is this? (4)

Oxygen (2); fine grained saturated soil and little disturbance (1) causes any present oxygen to be used quickly by organisms, leaving it anoxic (1).

43. What is the critical zone? (3)

The surface and near surface of earth where water, air, rocks, and organisms interact (something to that effect)

44. What is the difference between a swamp and a marsh? What are at least two benefits to preserving these wetlands? (4)

Marshes are dominated by non-woody vegetation while swamps are predominantly woody vegetation (2); (1 point each for any two) flood protection, erosion control, species habitats, nutrient cycling. (2)

45. What is a pingo? What latitudes are they typically found in (relatively)? (3)

A hill formed by a core of water freezing and expanding under the permafrost; high (1)

46. What defines permafrost? How does permafrost thaw contribute to climate change? (4)

Soil that is frozen solid year round (2); Trapped methane and other greenhouse gases are released from the soil as warming temperatures allow it to thaw (2).

47. Why would groundwater in karst areas not be as safe to drink? (5)

Water in karst areas runs through caverns from the surface (2); as a result, pollutants, contaminants, and flood waste can be washed directly into the groundwater (2) without undergoing the filtering process that would be had in an aquifer (1).

48. What are five negative effects caused by groundwater depletion? (5)

Land subsidence (1), Reduced streamflow/surface water (1), lowered water quality (1), increased pumping costs (1), lowered water table (1).

49. Is industry or agriculture a greater contributor to water pollution in the US today? Why might this be the case? (4)

Agriculture (2); Industry tends to be concentrated and is point source in nature (1), while agriculture is spread over a wide area of enforcement and is non point source from phenomena such as surface runoff (1)

50. Indonesia, the 4th largest country in the world by population, is currently planning to move its capital from the populous, sprawling coastal city of Jakarta on the island of Java to an inland location on the larger island of Borneo. This is due to the fact that Jakarta is currently sinking. What is the hydrological explanation for this? How could this have been prevented? (6)

An aquifer under the city (1) was being depleted (1) faster than it could recharge, causing the ground to be less supportive of structures/land subsidence (2) and leading to sinkage. This could have been prevented with reducing groundwater consumption (1) or limiting/removing the overdevelopment of the watershed surface (1)